LARK-1 System Requirements Specification (SAMPLE)

Doc. LARK-SRS-001, Issue 0.3 (draft) - SAMPLE DATA, fictional mission, synthetic content

SAMPLE DATA - FICTIONAL MISSION. This document is synthetic and was created to demonstrate ECSS Copilot. LARK-1 is not a real spacecraft, and the requirements contain deliberate flaws.

1 Scope: this System Requirements Specification (SRS) defines the system-level requirements for the LARK-1 3U CubeSat, an Earth-observation technology demonstrator.

2 Applicable documents: [AD-3] Launch provider ICD, [AD-4] Launch environment specification, [AD-5] Software coding standard.

3 Requirements

3.1 Mission and system

Req ID	Requirement text	Rationale
LARK-SYS-010	The LARK-1 spacecraft shall have a total mass not exceeding 4.0 kg including margins.	
LARK-SYS-020	The spacecraft shall fit within the 3U CubeSat envelope of 100 mm x 100 mm x 340.5 mm.	
LARK-SYS-030	The spacecraft shall operate for a mission lifetime of at least 2 years in a 520 km sun-synchronous orbit.	
LARK-SYS-040	The spacecraft shall be easy to integrate and shall be compatible with all commercial deployers.	
LARK-SYS-050	The spacecraft shall re-enter the atmosphere within 5 years after end of mission.	
LARK-SYS-060	The system should be designed with high reliability.	
LARK-SYS-070	The spacecraft shall comply with the launch provider interface control document [AD-3].	

3.2 Electrical power subsystem (EPS)

Req ID	Requirement text	Rationale
LARK-EPS-010	The EPS shall provide a regulated 5 V ± 0.1 V bus with a	

	maximum ripple of 50 mV peak-to-peak.	
LARK-EPS-020	The EPS shall provide adequate power to all subsystems in all modes.	
LARK-EPS-030	The battery shall have a capacity of 40 and shall be charged by the solar arrays.	
LARK-EPS-040	The EPS shall generate at least 7.5 W orbit-average power at end of life.	
LARK-EPS-050	The EPS shall switch the spacecraft to safe mode within 2 s when the battery voltage drops below 6.4 V.	
LARK-EPS-060	Shall protect the battery against overcharge and/or deep discharge.	
LARK-EPS-070	The EPS shall provide 3 switchable power lines to the payload, current limited to TBD mA.	

3.3 On-board computer and software (OBC)

Req ID	Requirement text	Rationale
LARK-OBC-010	The OBC shall boot into safe mode within 30 s after power-on.	
LARK-OBC-020	The on-board software shall be user-friendly and fast.	
LARK-OBC-030	The OBC shall store at least 2 GB of payload data.	
LARK-OBC-040	The on-board software shall never crash under any conditions.	
LARK-OBC-050	Telecommands shall be time-tagged and executed as appropriate.	
LARK-OBC-060	The OBC shall reject telecommands with an invalid CRC and shall increment the rejected-command counter by 1.	
LARK-OBC-070	The on-board software shall be written in C and comply with the project coding standard [AD-5].	
LARK-OBC-080	The watchdog shall reset the OBC if it is not serviced within 10.	

3.4 Communications (COM)

Req ID	Requirement text	Rationale
LARK-COM-010	The UHF transceiver shall transmit telemetry at a data rate of 9.6 kbps.	
LARK-COM-020	The S-band downlink shall provide a link margin of at least 3 dB at 10 deg elevation.	
LARK-COM-030	The communication system will support various ground stations, etc.	
LARK-COM-040	The UHF antenna shall deploy no earlier than 30 min after separation from the deployer.	
LARK-COM-050	The S-band transmitter shall not transmit before the ground has sent the TX-enable command.	
LARK-COM-060	The spacecraft shall maintain contact with the ground as far as possible.	

3.5 Attitude determination and control (ADCS)

Req ID	Requirement text	Rationale
LARK-ADC-010	The ADCS shall point the payload boresight with an accuracy better than 1 deg (3 sigma).	
LARK-ADC-020	The ADCS shall detumble the spacecraft from 10 deg/s to below 0.5 deg/s within 3 orbits.	
LARK-ADC-030	The ADCS shall provide sufficient pointing stability for imaging, TBC.	
LARK-ADC-040	The ADCS shall use a Kalman filter for attitude estimation.	

3.6 Payload, structure and environment

Req ID	Requirement text	Rationale
LARK-PLD-010	The payload camera shall acquire images with a ground sampling distance of 30 m from 520 km altitude.	
LARK-STR-010	The structure shall withstand the qualification random vibration levels of [AD-4] without damage.	
LARK-STR-020	The first natural frequency of the spacecraft shall be above 100 Hz.	
LARK-STR-030	All external surfaces shall be	

	labelled with the spacecraft serial number.	
LARK-THR-010	The thermal control shall keep the battery between 0 °C and 40 °C in all operational modes.	
LARK-THR-020	The thermal design shall keep it within limits.	
LARK-STR-040	The spacecraft shall be provided with a remove-before-flight pin accessible after integration into the deployer, it shall be colored red, and it shall be included in the delivered hardware with documentation describing how it is installed and removed by the integration team at the launch site.	

4 Requirements without ID (draft notes from the review)

The ground segment shall archive all telemetry. The mission control software should display alarms. The ground segment shall archive all telemetry.